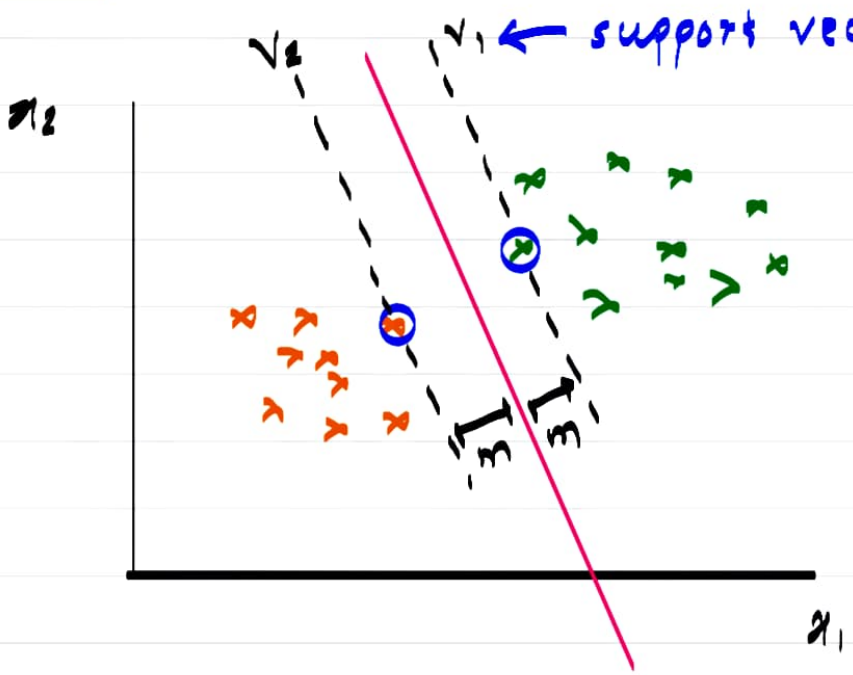
Machine Learning

27-05-26

SVM: Support Vector Machines

- Supervised ML, used for regression and classification
- parametric model

SVM classifier



SVM tries to find a decision boundary betⁿ classes and at the same time it tries to maximize margin betⁿ decision boundary and support vectors

Hypothesis function

$x \longrightarrow$ [SVM clf] $\longrightarrow y_{pred}$

eqⁿ of decision boundary

$$y = w_1x_1 + w_2x_2 + b = w_1x_1 + w_2x_2 + w_0$$

$$y = wx$$

$y = wx$ → decision boundary

if $wx < 0$ $\quad w_1 x_1 + w_2 x_2 + w_0 < 0$

then $y_{pred} = -1$ → -ve class

if $wx > 0$ $\quad w_1 x_1 + w_2 x_2 + w_0 > 0$

then $y_{pred} = 1$ → +ve class

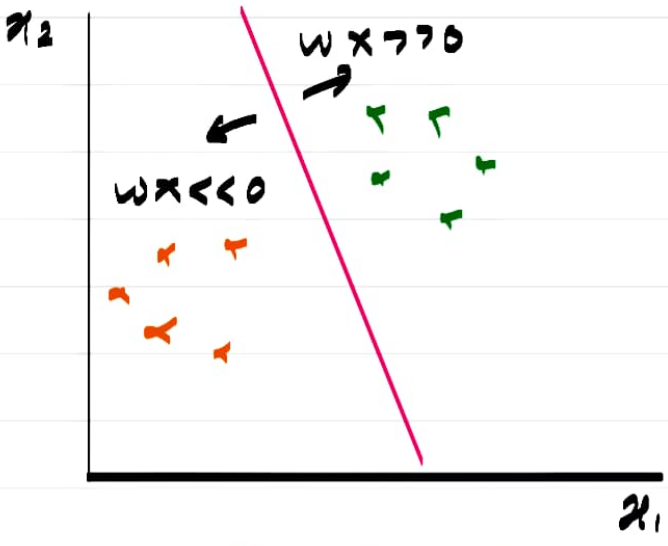
Loss Function:

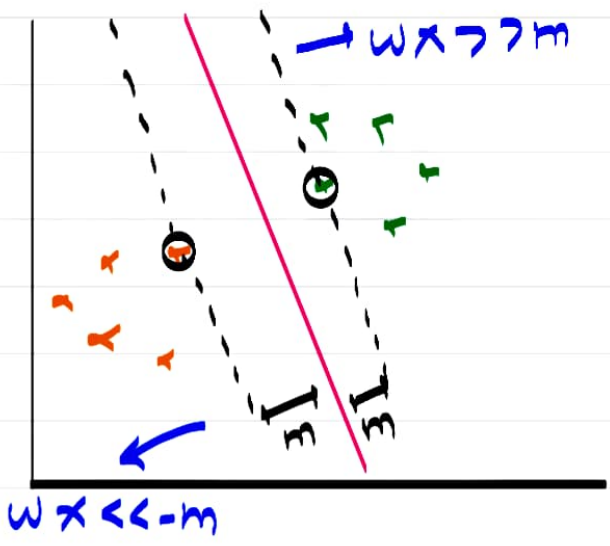
if $y_{act} = 1$ then we want

$y_{pred} >> m$

if $y_{act} = -1$ then we want

$y_{pred} << -m$





generally we use $m = 1$

with $m = 1$

if $y_{act} = 1$ then we want

$y_{pred} >> 1$

$wx >> 1$

if $y_{act} = -1$ then we want

$y_{pred} << -1$

$wx << -1$

} put a decision boundary in between with margin

Loss eqn satifying above requirements

$$L = \frac{1}{n} \sum_{i=1}^{n} \max[0, 1 - y_i(wx)]$$

Hinge Loss

n - no. of training datapoints

y_i - y_{act} for ith datapoint

When

$y_{act} = y_i = 1$

$wx_i = 2 = y_{pred}$

↓

model is doing good

↓

lesser loss

$L_i = \max[0, 1 - y_i wx_i]$

$= \max[0, 1 - 1 \times 2]$

$= \max[0, -1]$

$L_i = 0$ ↓

$y_{act} = y_i = 1$

$wx_i = 0.5 = y_{pred}$

↓

model is doing bad

↓

higher loss

$L_i = \max[0, 1 - y_i wx_i]$

$= \max[0, 1 - 1 \times 0.5]$

$= \max[0, 0.5]$

$L_i = 0.5$ ↑

- Objective of SVM classifier is to minimize the hinge loss

- It minimizes hinge loss using gradient descent

$$L = \frac{1}{n} \sum_{i=1}^{n} \max[0, 1 - y_i \underline{w} x_i]$$

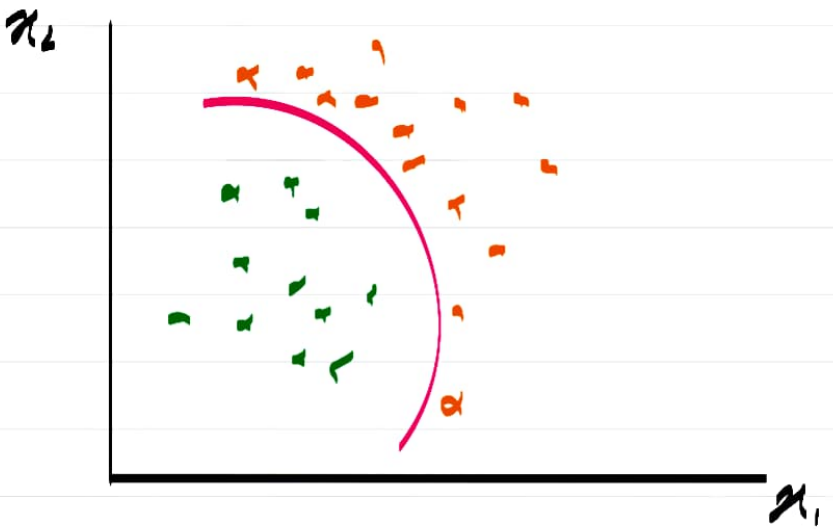
$\frac{\partial L}{\partial w} \Rightarrow$ gradients

update weights as

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

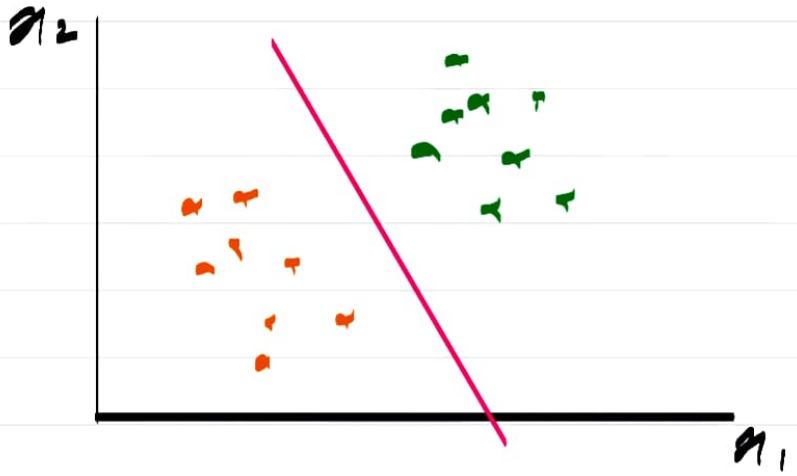
Kernel trick in SVM

- helps in training SVM with non-linear decision boundary



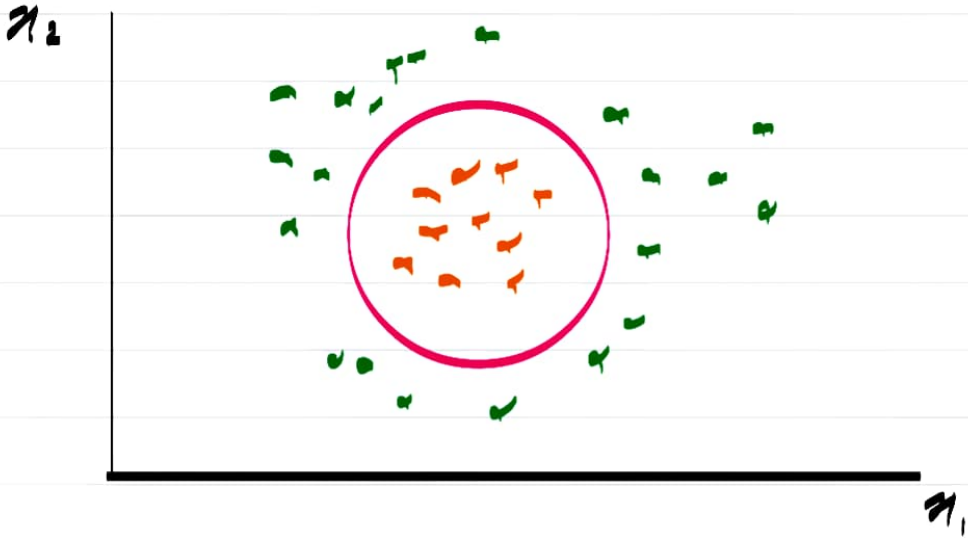
kernel = polynomial degree = 2

uses features like → x_1, x_2, x_1^2, x_2^2, $x_1 x_2$



Kernel = linear

uses features = x_1 x_2 x_1+x_2



Kernel = RBF Radial Basis function

uses features = x_1 x_2 x_1^2 x_2^2 $x_1^2+x_2^2$

- Kernel Type is a HP in SVM, needs tunning

SVM with Regularization

with regularization

$$L = C \times \frac{1}{n} \sum_{i=1}^{n} \max[0, 1 - Y_i w X_i] + \sum_{j=1}^{m} w_j^2$$

m - no. of weights

C - HP to control regularization

C ↑ → regularization ↓ → overfiting ↑

C ↓ → regularization ↑ → overfiting ↓

L1 regularization

$$L = cx\frac{1}{n}\sum_{i=1}^{n} \max[0, 1 - y_i w x_i] + \sum_{j=1}^{m} |w_j|$$